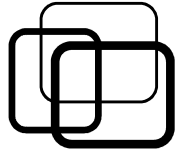


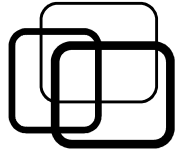
Text File

Inst. Nguyễn Minh Huy

Contents



■ Text file.



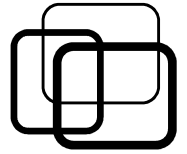
■ File IO:

■ Advantages (vs. keyboard and screen):

- Do not need user:
 - Automatic control.
 - Repeatable.
 - Communicate with other programs.
- Persistent storage.

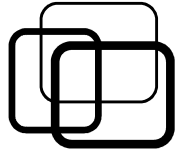
■ Disadvantages (vs. RAM):

- Sequential vs. random access.
- Slower.



■ File stream:

- File is either input or output device.
- File stream is connection between program and file.
- Declaration:
 - C (library <stdio.h>): **FILE** *<name>.
 - C++ (library <fstream>): **std::fstream** <name>.
- Steps to process a file:
 - Open file stream.
 - Read/write file stream.
 - Close file stream.



■ Open file stream:

■ C syntax:

- **fopen**("<file path>", "<open mode>");
- Return: file stream (succeeded) or NULL (error).

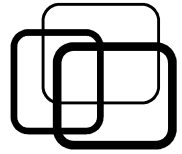
```
FILE *f = fopen("C:/test.txt", "r");  
if ( !f ) fprintf(stderr, "Error.");  
else printf("Succeeded.");
```

■ C++ syntax:

- **<stream>**("<file path>", <C++ open mode>);
- Check **<stream>** to know the result.

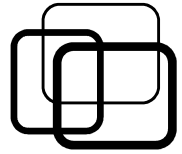
```
std::fstream f ("test.txt", std::ios::in);  
if ( !f ) std::cerr << "Error.;"  
else std::cout << "Succeeded.;"
```

Text file



- Open file stream:
 - C/C++ open modes:

Open mode C/C++	Ý nghĩa
r / std::ios::in	R ead-only, open for reading (text mode). Return NULL if file not found.
w / std::ios::out	W rite-only, open for writing (text mode). File created or overwritten.
a / std::ios::app	A ppend-only, open for append (text mode). File created if not found.
[r/w/a]+ / combine	Combine read and write (text mode).
[r/w/a]b / combine	Read and write in binary (binary mode).



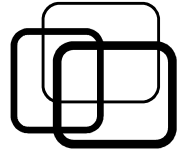
■ Close file stream:

■ C syntax:

- **fclose(<stream>);**
- Close an opened stream.
- Always close a stream after used.

```
FILE *f = fopen("C:\\test.txt", "r");
if ( !f ) {
    fprintf(stderr, "Error.");
    fclose( f );          // Wrong.
} else {
    printf("Succeeded.");
    fclose( f );          // Right.
}
```

■ C++ syntax: <stream>.**close();**



■ Read stream:

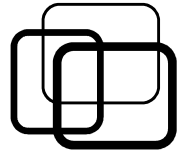
■ C syntax:

- **fscanf**(<stream>, "<type format>", &<var 1>, ...);
- **fgets**(<string>, <max line>, <stream>);

```
FILE *f = fopen("C:\\test.txt", "r");
if ( !f ) fprintf(stderr, "Error.");
else {
    int  a, b;      char s [ 100 ];
    fscanf( f, "%d %d", &a, &b );
    fgets( s, 100, f );
    fclose( f );
}
```

■ C++ syntax:

- <stream> >> <var>.
- <stream>.**getline**(<string>, <max line>).



■ Write stream:

■ C syntax:

➤ **fprintf**(<File stream>, "<Định dạng kiểu>", <Biến 1>, ...);

```
FILE *f = fopen("C:\\BaiTap.txt", "w");
```

```
if ( !f ) fprintf(stderr, "Error.");
```

```
else {
```

```
    int    a = 12;
```

```
    float  b = 4.165;
```

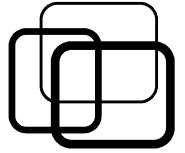
```
    char   c = 'A';
```

```
    fprintf( f, "Gia tri = %d %f %c", a, b, c );
```

```
    fclose( f );
```

```
}
```

■ C++ syntax: <stream> << <var>.

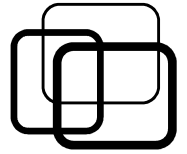


■ File management:

■ Delete file:

- Syntax: **remove**("<file path>");
- Return: 0 (succeeded), -1 (error).
- Do not need file stream.

```
if (remove("C:\\test.txt") == 0)
    printf("File deleted.\n");
else
    printf("Error: cannot delete file.\n");
```



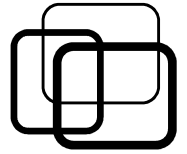
■ File management:

■ Rename/move file:

- Syntax: **rename**("<file path>", "<new file path>");
- Return: 0 (succeeded), -1 (error).
- Both path must be in the same drive.
- Do not need file stream.

```
if (rename("C:/test.txt", "C:/folder\\test2.txt") == 0)
    printf("Succeeded.\n");
else
    printf("Error.\n");
```

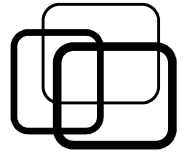
Summary



■ Text file:

- File stream: connection between program and file.
- C file stream: `FILE *` (`<stdio.h>`).
 - Open/close: `fopen`, `fclose`.
 - Read/write: `fscanf`, `fprintf`, `fgets`.
- C++ file stream: `std::fstream` (`<fstream>`).
 - Open/close: `<stream>()`, `<stream>.close`.
 - Read/write: `<stream> >>`, `<<`, `<stream>.getline`.





■ Practice 6.1:

File MATRIX.TXT stores a matrix of $M \times N$ integers:

- First line: M N (integers, rows and columns).
- Next M lines, line i stores N elements at row i of matrix.

Write C/C++ program to:

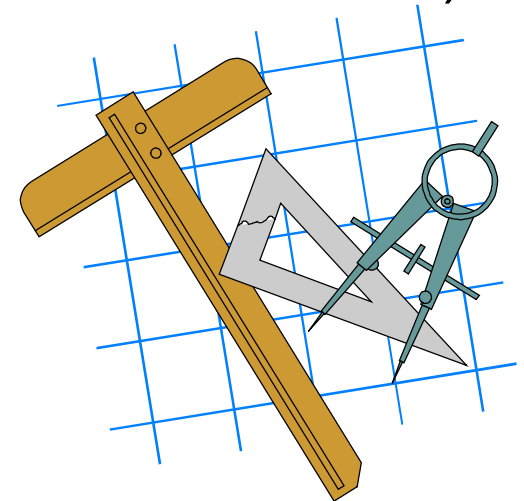
- Read matrix from MATRIX.TXT.
- Rotate left the matrix.
- Write the result to OUTPUT.TXT (same format as MATRIX.TXT).

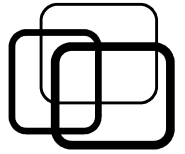
MATRIX.TXT

```
2 3
1 2 3
4 5 6
```

OUTPUT.TXT

```
3 2
3 6
2 5
1 4
```





■ Practice 6.2:

File NUMBER.TXT stores sequence of integers separated by space.

Write C/C++ program as follow:

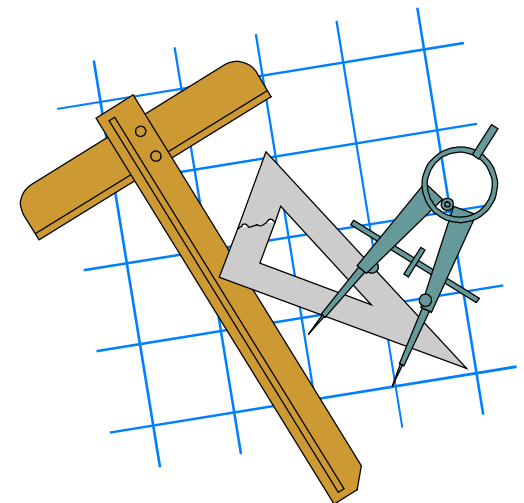
- Read sequence of integers from NUMBER.TXT.
- Extract prime numbers from the sequence.
- Write result to PRIME.TXT (same format as NUMBER.TXT).

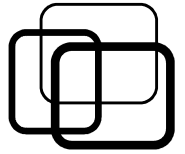
NUMBER.TXT

5 1 2 9 46 23 11 39 43 117 20

PRIME.TXT

5 2 23 11 43





■ Practice 6.3:

File OPERATION.TXT stores a list of arithmetic operations, each operation is on a line containing an operator (character +, -, *, /) in between two operands (integers).

Write C/C++ program as follow:

- Read list of operations from OPERATION.TXT.
- Compute the result of each operation.
- Write operation results to RESULT.TXT.

OPERATION.TXT

12 + 5

2 * 31

3 / 0

60 / 2

7 - 20

RESULT.TXT

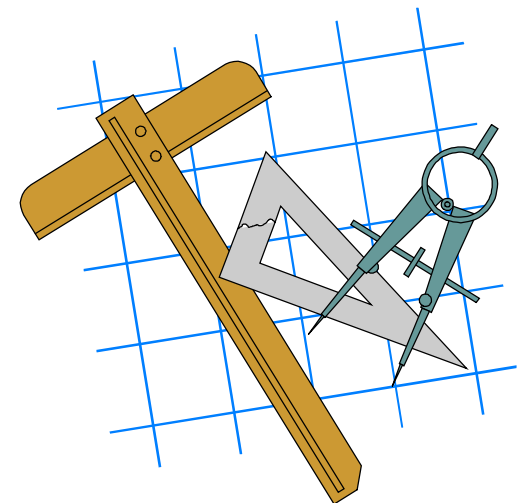
17

62

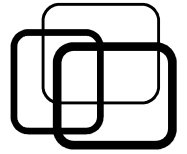
Divided by zero

30

-13



Practice



■ Practice 6.4:

File STUDENT.TXT stores a list of students, each student is on 2 lines:

- First line format: student name|student id.
- Second line format: student points (floats) separated by spaces.

Write C/C++ program as follow:

- Read student list from STUDENT.TXT.
- For each student, capitalize student name and compute GPA.
- Write the result to GPA.TXT in descending order of GPA.

STUDENT.TXT

nguyen van a|24127001

8.5 6.0 7.5

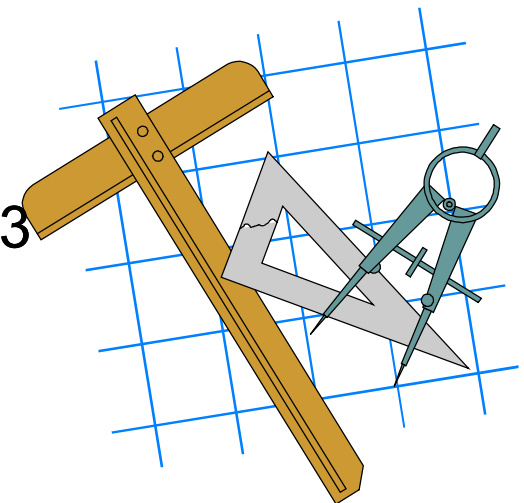
tran thi b|24127002

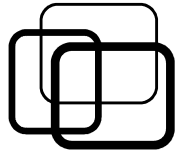
9.0 8.0 6.5 7.5

GPA.TXT

24127002|Tran Thi B|7.6

24127001|Nguyen Van A|7.3





■ Practice 6.5:

Write C/C++ program to count words and numbers:

- Read from a text file INPUT.TXT.
- Count and print the number of words and numbers in the file.

Notes:

- A word is a sequence of alphabets.
- A number is a sequence of digits.

INPUT.TXT

-----The, quick ##123 fox,,, 456!!!

Screen output:

Word count: 3.

Number count: 2.

